Smart Air Quality Monitoring and Alert System

BCSE301L — Software Engineering

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Project Overview

Introduction

A comprehensive software solution for real-time environmental monitoring, data analysis, and instant alert generation to safeguard public health.

Societal Problem

Health risks including respiratory and cardiovascular diseases due to lack of real-time pollution data and immediate alert mechanisms.

Target Users

Citizens

Municipal Authorities

Healthcare Organizations

Citizens

Real-time AQI data & health alerts

Municipal Auth.

Pollution trends & policy insights

Healthcare

Patient risk assessment data

Project Objectives

High Accuracy Data Collection

Deploy calibrated sensors capable of measuring PM2.5, PM10, CO, NO_2, SO_2, and O_3 with ≥95% accuracy.

Minimal Latency Processing

Process and transmit sensor data with latency under 5 seconds for near real-time monitoring.

Instant Alert Generation

Automatically trigger SMS, email, and push notifications when AQI exceeds defined thresholds.

Reliable Data Storage

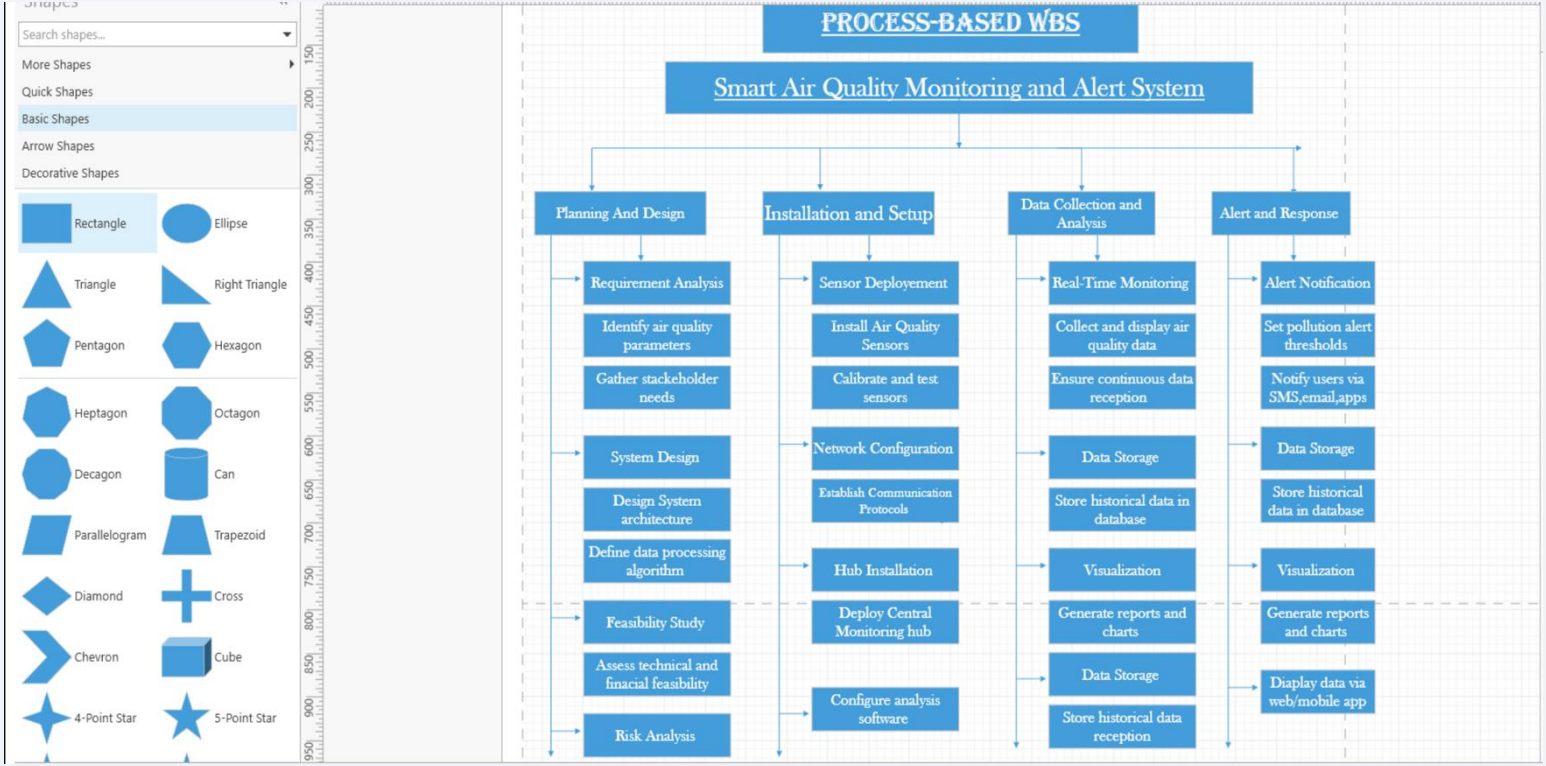
Store historical air quality data with 99.9% uptime for trend analysis and reporting.

User-Friendly Dashboard

Provide web and mobile interfaces with intuitive AQI visualizations, graphs, and health advisories.

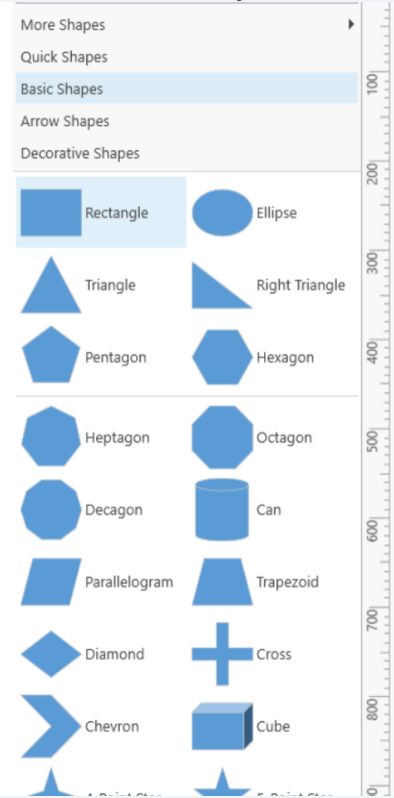
Project Planning — WBS (Process-Based)

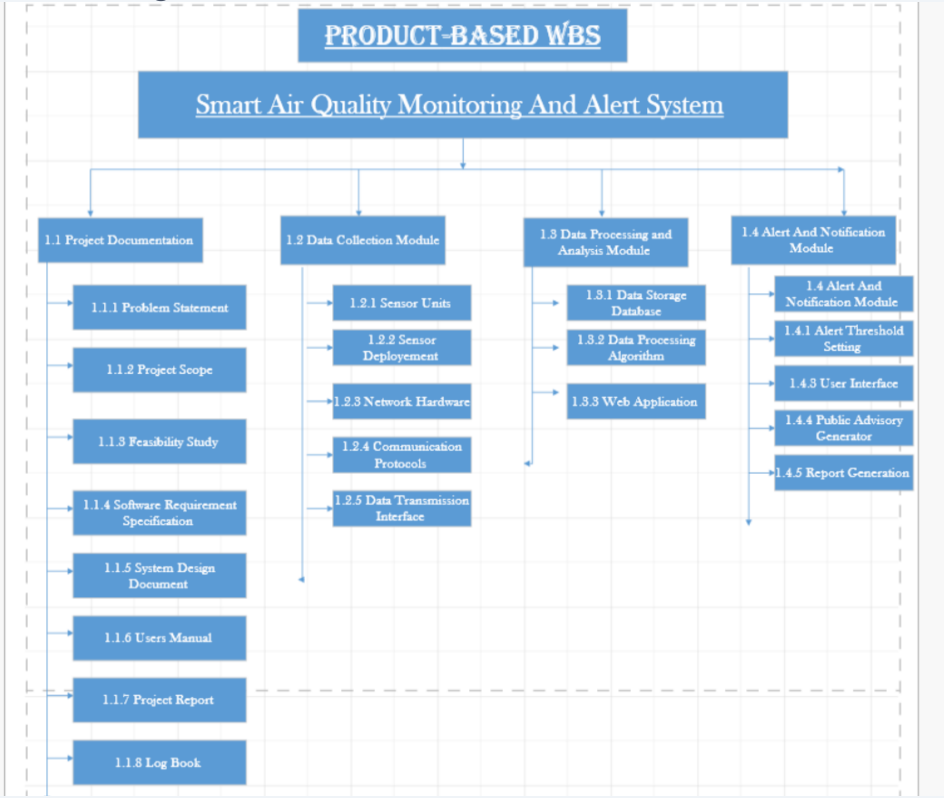
Phases: Planning & Design → Installation & Setup → Data Collection & Analysis → Alert & Response



Project Planning — WBS (Product-Based)

Deliverables: Project Documentation, Sensor Nodes, Processing Module, Notification Module

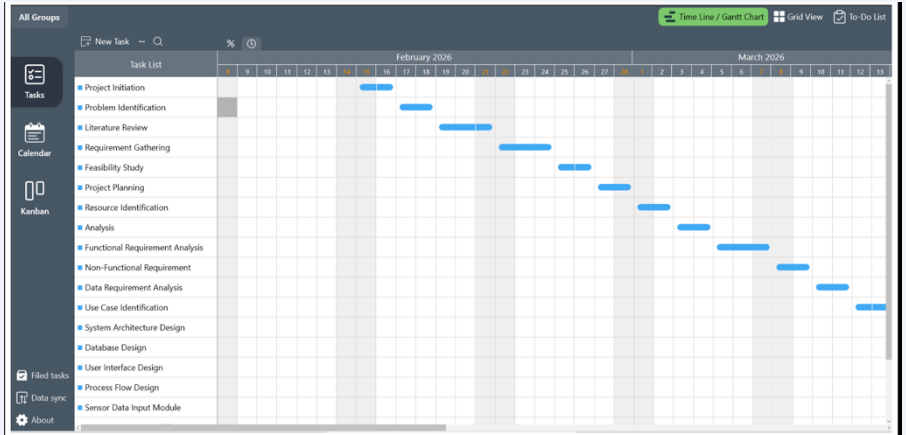


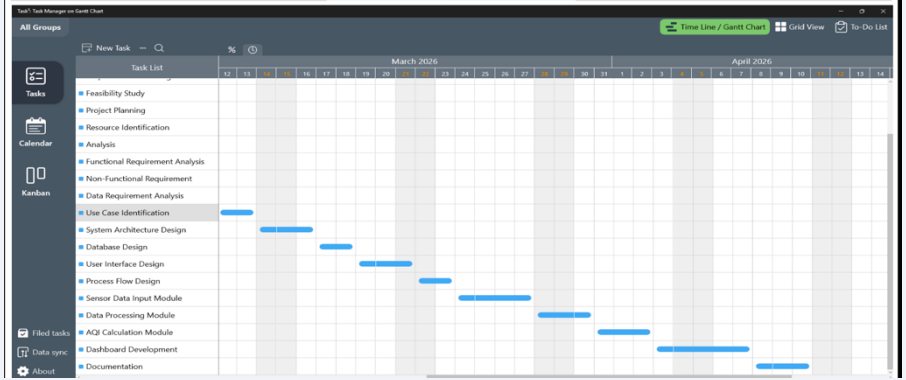


Project Schedule — Gantt Chart (Process-Based)

Timeline for process-based tasks from February 2026 to April 2026.

Task No	Activity	Start Date	End Date
1	Project Initiation	15-02-2026	16-02-2026
2	Problem Identification	17-02-2026	18-02-2026
3	Literature Review	19-02-2026	21-02-2026
4	Requirement Gathering	22-02-2026	24-02-2026
5	Feasibility Study	25-02-2026	26-02-2026
6	Project Planning	27-02-2026	28-02-2026
7	Resource Identification	01-03-2026	02-03-2026
8	Analysis	03-03-2026	04-03-2026
9	Functional Requirement Analysis	05-03-2026	07-03-2026
10	Non-Functional Requirement	08-03-2026	09-03-2026
11	Data Requirement Analysis	10-03-2026	11-03-2026
12	Use Case Identification	12-03-2026	13-03-2026
13	System Architecture Design	14-03-2026	16-03-2026
14	Database Design	17-03-2026	18-03-2026
15	User Interface Design	19-03-2026	21-03-2026
16	Process Flow Design	22-03-2026	23-03-2026
17	Sensor Data Input Module	24-03-2026	27-03-2026
18	Data Processing Module	28-03-2026	30-03-2026
19	AQI Calculation Module	31-03-2026	02-04-2026
20	Dashboard Development	03-04-2026	07-04-2026
21	Documentation	08-04-2026	10-04-2026

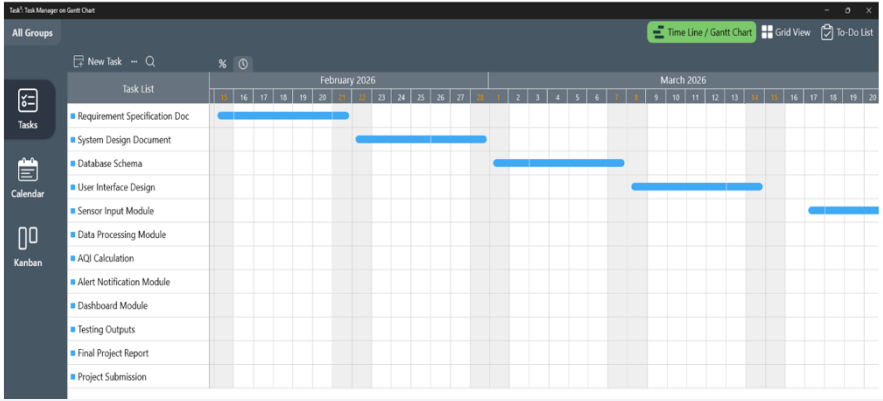


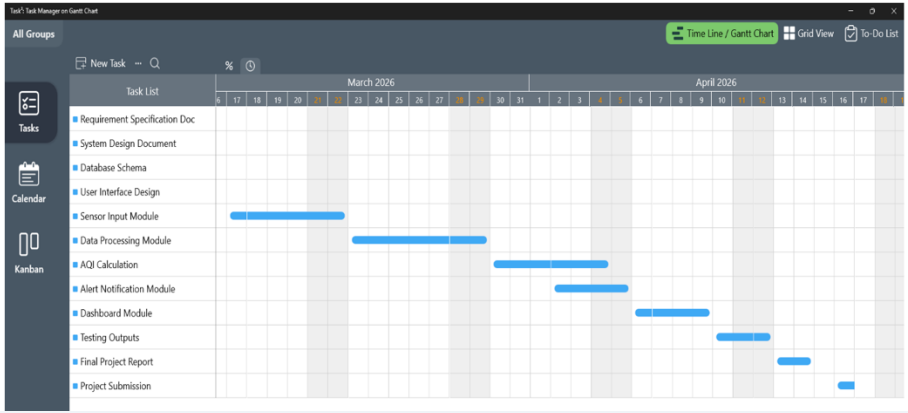


Project Schedule — Gantt Chart (Product-Based)

Timeline for product-based deliverables.

	A	B	C	D
1	TASK NO	TASK NAME	START DATE	END DATE
2	1	Requirement Specification Doc	15-02-2026	21-02-2026
3	2	System Design Document	22-02-2026	28-02-2026
4	3	Database Schema	01-03-2026	07-03-2026
5	4	User Interface Design	08-03-2026	14-03-2026
6	5	Sensor Input Module	17-03-2026	22-03-2026
7	6	Data Processing Module	23-03-2026	29-03-2026
8	7	AQI Calculation	30-03-2026	04-04-2026
9	8	Alert Notification Module	02-04-2026	05-04-2026
10	9	Dashboard Module	06-04-2026	09-04-2026
11	10	Testing Outputs	10-04-2026	12-04-2026
12	11	Final Project Report	13-04-2026	15-04-2026
13	12	Project Submission	16-04-2026	16-04-2026





Requirement Elicitation

Technique Used

Observation

Direct observation of environmental conditions in urban and industrial areas to understand real-world pollution exposure patterns and citizen behavior.

Key Finding

Citizens face elevated health risks due to:

- Absence of real-time air quality alerts
- No localized pollution data availability
- Delayed government advisories
- Lack of personal exposure tracking

Viewpoint-Oriented Analysis

Citizen Viewpoint

- Real-time AQI display
- Health risk notifications
- Location-based air quality data
- Historical trend access

Environmental Authority

- City-wide AQI analytics
- Pollution source identification
- Trend analysis & forecasting
- Regulatory compliance data

System Administrator

- Sensor health monitoring
- System uptime & maintenance
- User management
- Database backup & recovery

System Design Overview

Design Approach

The system follows a modular, layered architecture separating concerns across presentation, business logic, persistence, and database layers. UML diagrams were created to model structure and behavior.

Tools Used

Modelio 5.4
UML class, use case, and deployment diagrams

Draw.io
Architecture diagrams and WBS charts

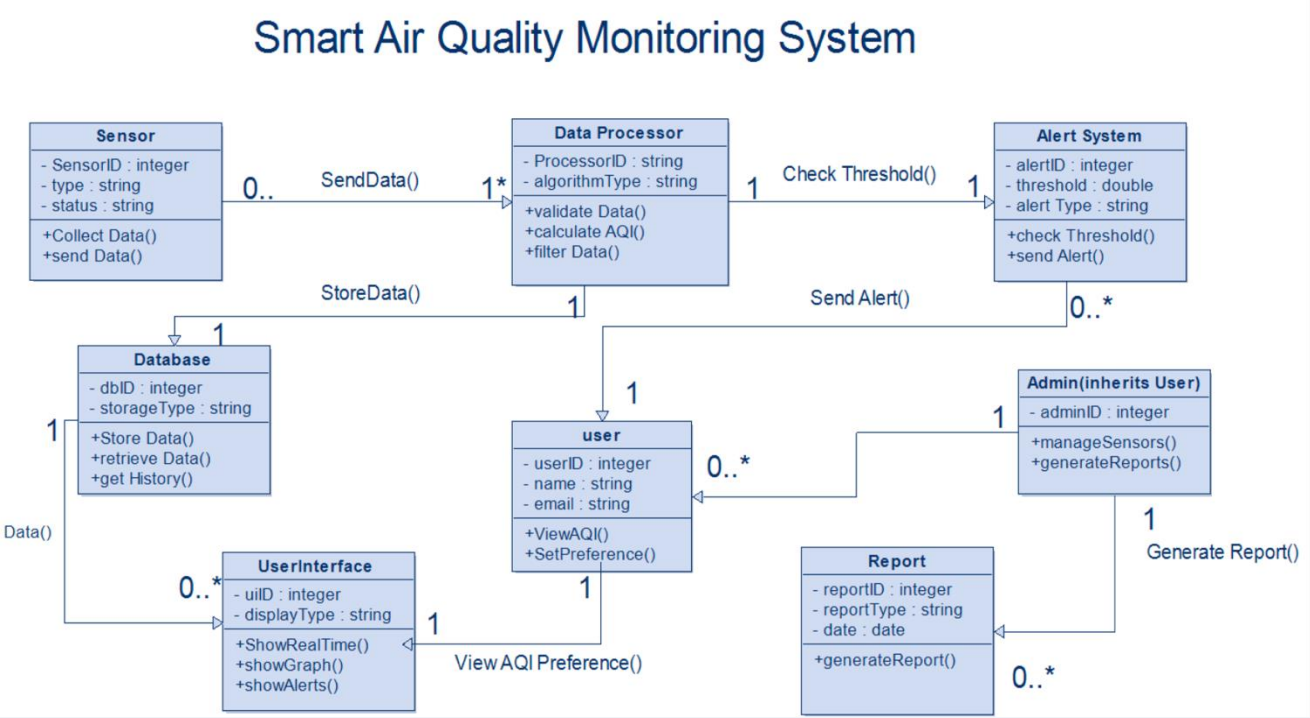
MS Project
Gantt charts and project scheduling

Design Principles

- Modularity & Separation of Concerns
- High Cohesion within modules
- Low Coupling between modules
- Scalable layered architecture
- Real-time data pipeline design

Detailed Design — Class Diagram

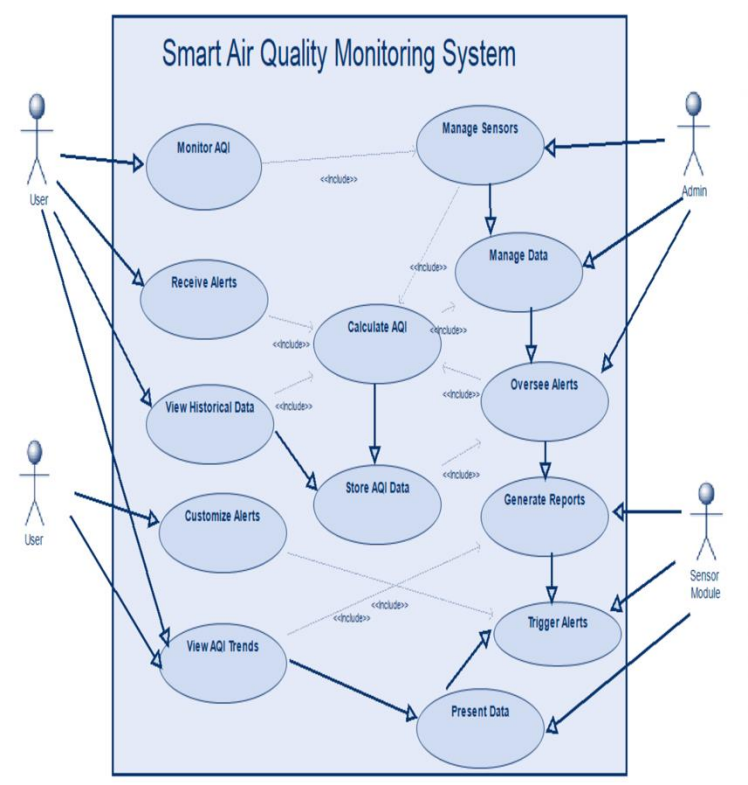
Classes: Sensor, Data Processor, Alert System, Database, User, User Interface, Report



Detailed Design — Use Case Diagram

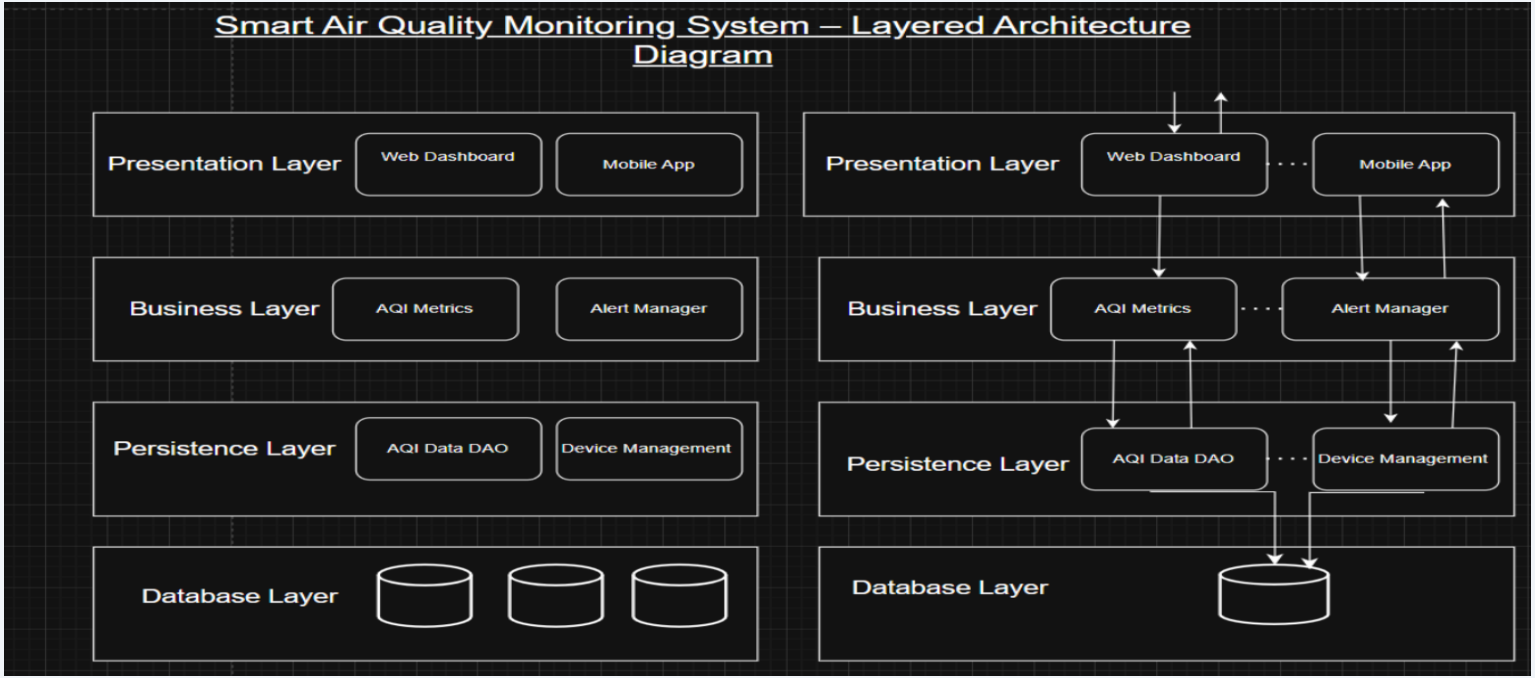
Use Case Actors
Citizen (User)
Environmental Authority
System Administrator

Core Functionalities
Monitor Real-Time AQI
Receive Instant Notifications
Evaluate System Performance



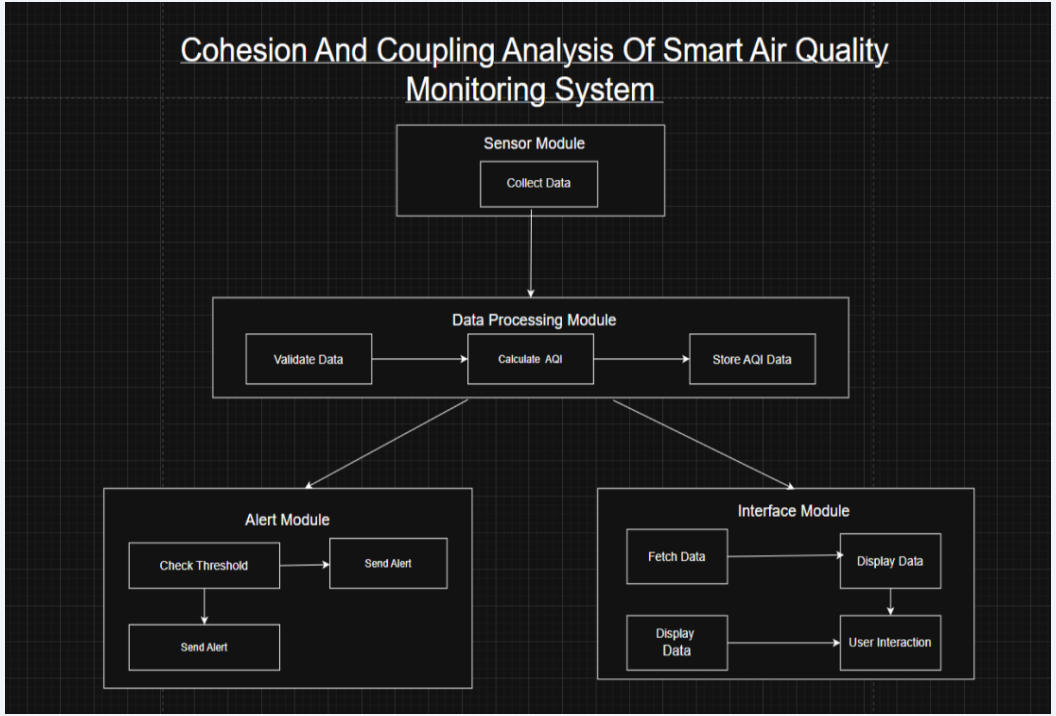
Architectural Design — Layered Architecture

Layers: Presentation → Business → Persistence → Database



Cohesion and Coupling Analysis

High Functional Cohesion | Low Data Coupling



Testing Strategy

Techniques

- Boundary Value Analysis (BVA)
- Equivalence Partitioning (EP)

100%

Pass Rate (6/6 TCs)

Test Case	Description	Technique	Result
TC-01	AQI boundary at 0	BVA	Pass ✓
TC-02	AQI boundary at 500	BVA	Pass ✓
TC-03	Valid pollutant range	EP	Pass ✓
TC-04	Invalid pollutant input	EP	Pass ✓
TC-05	Alert threshold trigger	BVA	Pass ✓
TC-06	Notification delivery	EP	Pass ✓

Conclusion & Future Scope

Summary

Successfully developed a modular, scalable framework for real-time Air Quality Index monitoring with automated alert generation, data visualization, and multi-user support.

Future Scope

- AI-based predictive pollution analysis
- Smart city IoT integration
- Machine learning for anomaly detection
- Cross-city AQI comparison dashboards

Thank You!

Questions & Discussion

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